

CSMFAB Materials Study — Part III

Specialty Functional Materials: Pigments, Coatings, Fluids, and Metamaterials

Carrington Storm Motors | Safe Pod Engineering Company | June 2026

Study Scope: YInMn Blue, CoAl₂O₄, MXene, AgNW, MR fluids, MR elastomers, shear-thickening fluid, WO₃ electrochromic material, and aerogel-derived specialty coatings.

Material 15 — YInMn Blue Pigment (Yttrium Indium Manganese Oxide)

Formula: YIn_{0.7}Mn_{0.3}O₃ — nominal | **Discovery:** Oregon State University, 2009 (M. Smith / Mas Subramanian) **Key property:** NIR-reflective “cool” blue; SRI=115; non-toxic; inorganic stability

15.1 Why YInMn Blue is the Most Important CSMFAB Surface Material

Used on **35 of 43 CSMFAB products** as primary exterior coating. Its unique crystal field absorption (Mn³⁺ in trigonal bipyramidal coordination, ~590 nm orange absorption) gives: - 85-92% NIR reflectance (700–2500 nm) - SRI=115 (vs. 10-25 for standard dark colors) - Temperature differential: up to 70°C cooler than dark surfaces under equivalent sun/fire exposure - Complete chemical stability (inorganic, no UV fade)

15.2 Chemical Composition and Precursors

Synthesis formula (solid-state method): Y₂O₃ + In₂O₃ + Mn₂O₃ → YInMnO₃ (calcined at 1200-1300°C in air)

Precursor elements and sources:

Yttrium (Y): - Rare earth element (see Part V for full supply chain) - From bastnäsite or monazite ore → rare earth separation → Y₂O₃ - Commercial Y₂O₃: \$35–\$55/kg

Indium (In): - CRITICAL MINERAL — primarily recovered as by-product of zinc smelting - Zinc ore deposits: China (Yunnan), Canada, South Korea, Peru - Price extremely volatile: \$150–\$350/kg (2026); historically \$80–\$800/kg range - Annual world production: ~700 metric tons — very limited - This is the portfolio’s single most supply-chain-vulnerable critical mineral

Manganese (Mn): - Abundant: South Africa (#1), Australia, Brazil, Ukraine - Mn₂O₃ or MnCO₃ precursor: \$1.20–\$2.50/kg — cheap and abundant! - No supply risk

15.3 YInMn Synthesis Routes

Route A — Solid-state calcination (current commercial standard): Mix Y₂O₃ + In₂O₃ + Mn₂O₃ in stoichiometric ratio → ball-mill blend → press pellets → fire at 1200°C, 12h

Route B — Hydrothermal synthesis (V2.0 advance, 2025): Y(NO₃)₃ + In(NO₃)₃ + Mn(NO₃)₂ dissolved in water → add NaOH → coprecipitate hydroxides → hydrothermal reactor at 250°C, 24h → wash → calcine at 900°C - Key advantage: lower calcination temperature (900°C vs. 1300°C) → lower energy - Equipment: autoclave reactor (\$8,000–\$25,000, 10L capacity)

Route C — Indium-sparing formulas (V2.0 critical research priority): Partial substitution of Ga, Fe, or Al for In → reduces Indium content by 30-60% Formula: Y(In_{0.4}Ga_{0.3}Mn_{0.3})O₃ → still blue-range, 70-80% NIR reflectance at 40% less In Formulation under development — potential to reduce In cost sensitivity dramatically

15.4 Commercial Acquisition

Supplier	Price/kg (2026)	Form
Shepherd Color Company (Cincinnati, OH)	\$75–\$100/kg	Standard grade
The Shepherd Color Company (high-purity)	\$110–\$145/kg	Glaze-grade
Ferro Corporation	\$80–\$110/kg	Industrial coating
BASF (Dispersions)	\$95–\$130/kg	Polymer-compatible
Custom synthesis (lab scale)	\$120–\$200/kg	Any university chem lab

Volume pricing at 3,200 kg/yr (portfolio annual): \$48–\$65/kg At 50,000 kg/yr (full scale): projected \$30–\$42/kg

15.5 In-House Synthesis — STRONGLY RECOMMENDED

Equipment for hydrothermal route: - Autoclave reaction vessels (2× 10L): \$20,000 - Ball mill for precursor mixing: \$5,000 - Calcination kiln (1100°C): \$12,000 (shared with ceramics program) - Particle characterization: \$8,000

In-house cost breakdown: - Y₂O₃: \$45/kg × 0.52 kg per kg YInMn = \$23.40 - In₂O₃: \$200/kg (refined In oxide, not metal) × 0.62 kg per kg YInMn = \$124.00 - Mn₂O₃: \$2.50/kg × 0.18 kg = \$0.45 - Energy + labor: \$18/kg - **Total in-house: ~\$165/kg (In-limited) vs. \$75–\$100 commercial**

Wait — commercial is CHEAPER currently? Yes, because Shepherd Color has In supply contracts at \$120–\$140/kg (vs. spot \$200–\$350/kg for oxide). Their volume leverage is not yet matchable.

In-house strategy: Research indium-sparing formulas aggressively. With 40% less indium: In-house cost falls to **\$85–100/kg** — now competitive with buy price. **Feasibility Rating: 3/5** for standard formula; **4/5 for indium-sparing R&D version**

15.6 Indium Recovery via Ionic Liquid — Phoenix Protocol Key Step

From used YInMn glazes on retired Aegis products: - H₂SO₄ leach □ [P8888][Cl] ionic liquid extraction □ In recovery 99.4% - Recovered In cost-basis: \$18–\$28/kg processing (vs. \$200–\$350/kg virgin) - **This makes Phoenix Protocol YInMn recycling economically dominant** - Yttrium also recovered: \$38 processing vs. \$35–\$55/kg virgin Y₂O₃

Products: 001–021, 024, 026–040, 050, 101, 105, CSMMECH 001

Material 16 — Cobalt Aluminate Spinel (CoAl₂O₄)

Formula: CoAl₂O₄ | **Color:** Deep blue (complementary to YInMn) | **Class:** Spinel structure oxide

16.1 Properties

- SRI: 98 (slightly below YInMn's 115 but still excellent)
- NIR reflectance: 70–80%
- Max service: >1400°C (higher than YInMn's ~1200°C limit)
- Contains no Indium — no critical mineral vulnerability
- Absorbs 1060 nm Nd:YAG wavelength selectively — laser countermeasure function
- Cost: significantly lower than YInMn

16.2 Synthesis

Spinel synthesis (co-precipitation or sol-gel): $\text{Co}(\text{NO}_3)_2 + \text{Al}(\text{NO}_3)_3$ in aqueous solution $\rightarrow$ add $\text{NH}_3 \rightarrow$ Co-Al hydroxide co-precipitate $\rightarrow$ calcine at 800-1000°C $\rightarrow$ CoAl_2O_4 spinel

Precursors: - $\text{Co}(\text{NO}_3)_2$ or CoSO_4 : cobalt from DRC/Congo mining; price \$28–\$45/kg Co metal equivalent
- $\text{Al}(\text{NO}_3)_3$: from alumina \$2/kg $\rightarrow$ nitrate conversion is straightforward - NH_3 : commodity

16.3 Commercial Pricing

- Standard ceramic grade: \$55–\$85/kg
- Pigment grade (fine, heat-stable): \$70–\$100/kg
- Heubach (Germany), Ferro Corp., Shepherd Color all supply

16.4 In-House Synthesis

Co-precipitation is simple bench chemistry. Equipment: pH-controlled reactor \$5,000, calcination furnace (shared). In-house cost: **\$22–\$38/kg** (vs. \$55–\$100 commercial) **Feasibility Rating:** 5/5 — simplest functional coating synthesis in portfolio.

Products: 001, 010, 011, 016, 021 (interior walls/Schumann surfaces), CSMIC 105 cross-arms

Material 17 — MXene ($\text{Ti}_3\text{C}_2\text{T}_x$) Electromagnetic Shielding

Full chemical name: $\text{Ti}_3\text{C}_2\text{T}_x$ where T = surface functional groups (OH, O, F) **Class:** 2D transition metal carbide | **Discovery:** Drexel University, 2011

17.1 The New Crown Jewel of EMI Shielding

SE = 92 dB at 45 μm thickness (2024) — surpasses all other flexible EMI materials. Used in 18 CSMFAB products; currently 2026's most expensive portfolio material at \$2,000/kg.

But: Price projected to fall 70-80% by 2028-2030 as production scales.

17.2 Synthesis Chemistry

Step 1 — MAX Phase precursor (Ti_3AlC_2): (same as Material 6) **Step 2 — Selective etching of Al layer:**

Method A (LiF/HCl — minimally intensive fluorine system): $\text{Ti}_3\text{AlC}_2 + \text{LiF} + \text{HCl}$ (12M) at 35°C for 24h $\rightarrow$ $\text{Ti}_3\text{C}_2\text{T}_x$ multilayer clay

Method B (HF etching — original Drexel method): $\text{Ti}_3\text{AlC}_2 +$ concentrated HF (48%) $\rightarrow$ $\text{Ti}_3\text{C}_2\text{T}_x$ *Fluoride waste stream requires neutralization treatment*

Method C (TMAOH — non-fluorine, 2024):** $\text{Ti}_3\text{AlC}_2 +$ tetramethylammonium hydroxide $\rightarrow$ $\text{Ti}_3\text{C}_2\text{T}_x$ (emerging; avoids HF)

Step 3 — Delamination: Multilayer clay + sonication in water $\rightarrow$ single-layer/few-layer MXene colloidal solution

Step 4 — Spray/filter into film or coating: - Sprayed onto substrate: 50-100 μm $\rightarrow$ 75+ dB SE - Vacuum-filtered membrane: freestanding film

17.3 Raw Materials for MXene Synthesis

- Ti_3AlC_2 MAX phase: \$80–\$420/kg commercial (or in-house \$35-55/kg — see Material 6)
- LiF: \$8–\$15/kg (lithium fluoride salt)
- HCl (12M): \$0.30/kg bulk hydrochloric acid

- TMAOH (tetramethylammonium hydroxide): \$25–\$45/kg
- DI water: \$0.001/L (in-house RO system)

From in-house MAX phase at \$35–\$55/kg: Conversion yield: ~80% ($\text{Ti}_3\text{AlC}_2 \rightarrow \text{Ti}_3\text{C}_2\text{T}_x$ by mass after etching) Chemical costs per kg MXene: \$35–\$55 (MAX) + \$12 (LiF, HCl) + \$8 (delamination) = **\$55–\$75/kg**

This is the most transformative in-house opportunity in the portfolio: - Current commercial MXene: \$1,500–\$2,500/kg - In-house from in-house MAX phase: \$55–\$75/kg - **Savings: 96–97%**

17.4 Commercial Acquisition (Until In-House Scales)

Supplier	Form	Price/kg (2026)
Drexel MXene/Murata partnership	Colloidal suspension	\$2,000–\$2,500
American Elements	Multilayer powder	\$1,500–\$2,000
2D Materials Inc.	Research quality	\$1,800–\$2,800
Taiyo Yuden / TDK (Japan)	Industrial trial	\$800–\$1,200

17.5 MXene Tile Pattern — Minimizing Usage

Key design insight: MXene provides GHz EMI shielding even in **discontinuous tile patterns**. Islands of 3×3 cm MXene separated by 1mm gaps block GHz frequencies while maintaining GIC immunity. This design reduces MXene content by **40-60%** vs. continuous coating —directly proportional cost savings.

Example: CSMFAB 029 Neural-Grip Wheel Wrap: - Full coverage: 0.05 kg MXene × \$2,000 = \$100 - Tile pattern: 0.02 kg × \$2,000 = \$40 - At in-house price (\$65/kg): 0.02 kg = \$1.30 — effectively negligible

Feasibility Rating: 5/5 for in-house (with in-house MAX phase program already running)

Products: 001, 011, 019, 021, 024, 029, 030, 032, 034, 037, 038, 050, CSMMECH 001

Material 18 — Magnetorheological Fluid (LORD MRF-140CG)

Type: Iron particle suspension in hydrocarbon carrier | **Supplier monopoly:** LORD Corporation (Parker Hannifin)

18.1 Composition and Properties

- Carbonyl iron particles: 7-10 μm diameter, 80-85 vol.%
- Carrier: hydrocarbon oil (mineral or synthetic)
- Stabilizer: fumed silica, surfactants
- Yield stress: 80 kPa at 250 kA/m (2025 specification)
- Off-state viscosity: 0.28 Pa·s at 40°C
- Operating range: -40°C to +130°C

18.2 Making MR Fluid In-House

Carbonyl iron powder (CIP) synthesis: Iron pentacarbonyl ($\text{Fe}(\text{CO})_5$) vapor decomposition at 250-300°C $\rightarrow$ spherical carbonyl iron particles - $\text{Fe}(\text{CO})_5$ source: from iron + CO at high pressure in autoclave - High-pressure equipment (200 bar CO): \$85,000–\$150,000 - Safety: CO gas handling requires extreme precautions

Alternative CIP source: BASF produces CIP at \$8–\$15/kg (Carbonyl Iron Powder) — Purchase CIP from BASF, formulate fluid in-house

In-house MRF formulation from CIP: 1. BASF CIP (\$12/kg) + hydrocarbon oil (\$2/kg) + fumed silica (\$4/kg) + dispersant (\$8/kg) 2. Mix in high-shear mixer; test rheology 3. Fill into product bladders

18.3 Commercial vs. In-House

Option	Cost/kg	Quality
LORD MRF-140CG commercial	\$250–\$300/kg	Certified, guaranteed performance
In-house from BASF CIP	\$25–\$40/kg	Requires rheology qualification

In-house savings: 85-90% Feasibility Rating: 4/5 — straightforward formulation; quality matching requires R&D effort. Critical R&D task: Match LORD's specific particle size distribution (7-10 μm) and surface treatment.

Products: 024, 029, 030, 031, 035, MAG (Bio-Sync)

Material 19 — Shear-Thickening Fluid (STF) — Thorax-Calm Seatbelt

Composition: SiO₂ nanoparticles (200 nm) in polyethylene glycol (PEG-400)

19.1 How STF Works

At rest (low shear): particles dispersed, fluid flows normally ($\eta_0 = 0.8 \text{ Pa}\cdot\text{s}$) At impact (high shear $\dot{\gamma} > 200 \text{ s}^{-1}$): particles jam in hydroclusters η rises to 85+ Pa·s Effect: impact-activated stiffening — seatbelt cover becomes rigid pad in crash

19.2 Raw Materials

- **Fumed SiO₂ (200 nm):** Cabot Aerosil® 200, or Evonik Aerosil®
 - Commercial: \$5–\$8/kg
 - Volume (10t/yr): \$3.50–\$5/kg
 - In-house synthesis (Stöber process from TEOS): **\$1.80–\$3.00/kg** — saves 40-60%
- **PEG-400 (polyethylene glycol):** commodity chemical \$1.20–\$2.50/kg
 - Sourced from Dow Chemical, Clariant — no in-house benefit

STF formulation: 60 vol.% SiO₂ in PEG-400 $\square$ sonicate to disperse $\square$ fill and seal in TPU capsule

In-house STF total cost: \$12–\$18/kg (vs. no established commercial STF product for this use; custom-mixed to order)

Products: 033 (Thorax-Calm seatbelt), 029 (Neural-Grip inner layer variant)

Material 20 — WO₃ Electrochromic Film (Stellar-Tint Window)

Formula: WO₃ tungsten trioxide | **Application:** Variable-tint smart window film

20.1 Working Principle

WO₃ + xLi⁺ + xe[−] $\square$ Li_xWO₃ (blue-colored, absorbing) + Li⁺ removal (clear) - Electrical switching at 1-2 V drives lithium intercalation/deintercalation - Switching time: <3 seconds (2025 grade) - Cycle life: 50,000+ cycles (150-year daily use estimate)

20.2 Raw Materials

Tungsten source: - Primary mineral: **Scheelite (CaWO_4)** or **Wolframite ($(\text{Fe,Mn})\text{WO}_4$)** - Major producers: China (85% world supply), Russia, Canada, Vietnam - **Supply risk: HIGH** (China dominance) - WO_3 commercial: \$35–\$55/kg tungsten oxide

WO_3 synthesis: - Ammonium paratungstate (APT) $\rightarrow$ calcine at 600°C in air $\rightarrow \text{WO}_3$ - APT from ore processing $\rightarrow$ \$28–\$42/kg

Alternative: Hydrogen tungsten bronze (H_xWO_3) — electrodeposited WO_3 film - Dissolve WO_3 in $\text{H}_2\text{O}_2/\text{H}_2\text{O}$ $\rightarrow$ ammonium tungstate solution $\rightarrow$ spin-coat $\rightarrow$ anneal

20.3 Commercial Window Film Pricing

- WO_3 nanoparticle ink: \$250–\$450/kg
- Roll-to-roll coated film (LiPON/ WO_3 /PEDOT stack): \$35–\$65/ m^2 (at scale; 2026 emerging market)
- Sage Glass (Saint-Gobain): \$35–\$100/ m^2 (installed, not just film)

20.4 In-House Thin-Film Deposition

WO_3 electrochromic layer deposition via: - Sol-gel spin coating (simplest): dissolve APT in water $\rightarrow$ coat $\rightarrow$ bake - Physical vapor deposition (sputtering): higher quality, more capital

In-house sol-gel cost: \$8–\$15/ m^2 WO_3 layer (from APT reagent) Combined PEDOT:PSS + AgNW transparent conductor: \$12–\$25/ m^2 Total Stellar-Tint in-house: **\$35–\$55/ m^2** (vs. \$60–\$180/ m^2 commercial specialty) **Feasibility Rating: 3/5** — sol-gel coating is accessible; roll-to-roll requires \$150K+ equipment.

Products: 034 (Stellar-Tint window film)

Material 21 — PEDOT:PSS Transparent Conductor (Window Film)

Full name: Poly(3,4-ethylenedioxythiophene) polystyrene sulfonate **Function:** ITO-free transparent electrode for electrochromic and EMI shielding applications

21.1 Properties and Why It Replaces ITO

- Sheet resistance: 50–500 $\Omega/\square$ (controllable by formulation)
- Optical transmittance: 88–95%
- Solution-processable: spray, spin-coat, blade-coat, inkjet
- Acid treatment (H_2SO_4 , formic acid) can reduce R_s to 15 $\Omega/\square$ — nearly ITO-equivalent
- Contains no Indium (unlike ITO) — zero critical mineral vulnerability
- Mechanical flexibility: survives 1000 bend cycles at 3mm radius

21.2 Synthesis

- EDOT monomer + PSS counter-polymer $\rightarrow$ oxidative polymerization in water
- Heraeus Clevios™ brand; Ossila Ltd. (research grade)
- Commercial PEDOT:PSS dispersion: \$80–\$180/kg (at 1-2% solid concentration)
- Solid content basis: \$8,000–\$18,000/kg film — mitigated by very thin layer usage

21.3 AgNW (Silver Nanowire) Enhancement

- AgNW + PEDOT:PSS hybrid achieves: 92% transmittance + 35 dB SE at 1 GHz
- AgNW synthesis: silver nitrate + polyvinylpyrrolidone (PVP) in ethylene glycol at 170°C $\rightarrow$ Ag nanowires 50-200 nm diameter $\times$ 5-30 μm length
- Commercial AgNW ink: \$800–\$2,500/kg (silver content-dependent)

- In-house polyol synthesis: **\$120–\$250/kg** (from AgNO₃ \$580/kg × 30% Ag content in wire)

Products: 034 (Stellar-Tint), 011 (building window film)

Summary Table — Part III Specialty Functional Materials

Material	Commercial (\$/kg)	In-House Estimate	Savings	Feasibility
YInMn Blue	\$75–\$100/kg	\$85–\$100/kg*	0–15%	3/5
YInMn Blue (indium-sparing formula)	\$75–\$100/kg	\$45–\$65/kg	20–40%	4/5**
CoAl ₂ O ₄	\$55–\$100/kg	\$22–\$38/kg	55–70%	5/5
MXene Ti ₃ C ₂ Tx	\$1,500–\$2,500/kg	\$55–\$75/kg	96–97%	5/5***
MR Fluid (in-house from CIP)	\$250–\$300/kg	\$25–\$40/kg	85–90%	4/5
STF (SiO ₂ /PEG-400)	custom \$30–\$60/L	\$12–\$18/L	55–75%	5/5
WO ₃ film (sol-gel)	\$35–\$100/m ²	\$8–\$15/m ²	65–80%	3/5
PEDOT:PSS	\$80–\$180/kg	commercial only	—	1/5
AgNW ink	\$800–\$2,500/kg	\$120–\$250/kg	75–80%	4/5

YInMn in-house currently cost-disadvantaged due to Indium supply chain **Indium-sparing formula reduces In by 40% — R&D priority** Requires in-house MAX Phase program (see Part I) — synergistic investment

End of CSMFAB Materials Study Part III | Specialty Functional Materials Carrington Storm Motors | June 2026